

Solutions for Practice Examination 1
Mathematical Probability (STAT2003/STAT7003)
Created during Semester 1, 2026

Question 1. [10 marks]

Let X have PDF $f(x) = c x^3(1 - x)$ for $0 \leq x \leq 1$ (and $f(x) = 0$ otherwise), where c is a normalising constant.

- (a) Find the value of c .

[2 marks]

- (b) Identify this as a Beta distribution and state the parameters α and β . Confirm your value of c using the Beta function.

[2 marks]

- (c) Suppose U_1, U_2, U_3, U_4, U_5 are i.i.d. $\mathcal{U}(0, 1)$ random variables and $U_{(1)} \leq U_{(2)} \leq \dots \leq U_{(5)}$ are the order statistics. Which order statistic has the same distribution as X ? Justify your answer using the fact that $U_{(k)} \sim \text{Beta}(k, n - k + 1)$ for a sample of size n .

[3 marks]

- (d) Find the CDF $F(x)$ and set up the equation for the median.

[3 marks]

Solution:

$$(a) \int_0^1 c x^3(1 - x) dx = c \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = c \left(\frac{1}{4} - \frac{1}{5} \right) = c \cdot \frac{1}{20} = 1.$$

Therefore $c = 20$.

- (b) $f(x) = 20 x^3(1 - x) = 20 x^{4-1}(1 - x)^{2-1}$. This is $\text{Beta}(\alpha, \beta)$ with $\alpha = 4, \beta = 2$.

Confirmation: $B(4, 2) = \Gamma(4)\Gamma(2)/\Gamma(6) = 6 \cdot 1/120 = 1/20$, so $c = 1/B(4, 2) = 20$. ✓

- (c) For a sample of size $n = 5$, the k -th order statistic has distribution $\text{Beta}(k, 5 - k + 1) = \text{Beta}(k, 6 - k)$.

We need $\alpha = k = 4$ and $\beta = 6 - k = 2$. So X has the same distribution as $U_{(4)}$, the fourth order statistic from a sample of 5 uniform random variables.

$$(d) F(x) = \int_0^x 20 t^3(1 - t) dt = 20 \left[\frac{t^4}{4} - \frac{t^5}{5} \right]_0^x = 5x^4 - 4x^5, \quad 0 \leq x \leq 1.$$

Median m : solve $5m^4 - 4m^5 = 1/2$. Numerically, $m \approx 0.6862$.

Question 2. [10 marks]

Consider the following Python code that simulates a sequence of random variables:

```
x = 0
n = 0
while x != 3:
    u = random()
    if x == 0:
        if u < 0.5:
            x = 0
        elif u < 0.8:
            x = 1
        else:
            x = 2
    elif x == 1:
        if u < 0.6:
            x = 1
        else:
            x = 3
    else:
        if u < 0.4:
            x = 1
        elif u < 0.7:
            x = 2
        else:
            x = 3
    n += 1
print(n)
```

- (a) Write down the transition matrix $\mathbf{P}$ for this Markov chain on the state space $\{0, 1, 2, 3\}$, with rows and columns in that order. Identify any states that, once entered, the chain never leaves.

[3 marks]

- (b) If the chain starts in state 0 at time $n = 0$, what is the probability that it is in state 3 at time $n = 2$?

[2 marks]

- (c) Suppose the code is modified so that the initial value is $x = 1$ instead of $x = 0$. Let N be the random variable representing the value of n printed. What is the distribution of N ? Compute $\mathbb{E}[N]$ and $\mathbb{P}(N > 4)$.

[5 marks]

Note about Python: `random()` generates a $\mathcal{U}(0, 1)$ random variable.

Solution:

(a)

$$\mathbf{P} = \begin{pmatrix} 0.5 & 0.3 & 0.2 & 0 \\ 0 & 0.6 & 0 & 0.4 \\ 0 & 0.4 & 0.3 & 0.3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Once the chain enters state 3 it stays there ($P_{33} = 1$). The code runs until the chain reaches state 3, so n counts the number of transitions until that happens.

(b) To reach state 3 in exactly 2 steps from state 0, the chain must be in a state that can transition to 3 at step 1 (i.e. state 1 or 2), and then go to 3 at step 2. Note: state 0 cannot go directly to state 3.

$$\mathbb{P}(X_2 = 3 | X_0 = 0) = (\mathbf{P}^2)_{0,3}.$$

$$= P_{00}P_{03} + P_{01}P_{13} + P_{02}P_{23} + P_{03}P_{33}$$

$$= 0.5(0) + 0.3(0.4) + 0.2(0.3) + 0(1) = 0 + 0.12 + 0.06 = 0.18.$$

(c) Starting from state 1, at each step the chain either goes to state 3 (with probability 0.4) or stays at state 1 (with probability 0.6). These are the only two transitions from state 1. Each step is an independent trial with success probability $p = 0.4$, so $N \sim \text{Geom}(0.4)$.

$$\mathbb{E}[N] = 1/0.4 = 5/2 = 2.5.$$

$$\mathbb{P}(N > 4) = (1 - 0.4)^4 = 0.6^4 = 0.1296.$$

Question 3. [10 marks]

Let X be a non-negative random variable with $\mathbb{E}[X] = 12$ and $\text{Var}(X) = 9$.

- (a) Prove Markov's inequality from first principles: for any non-negative random variable Y and $a > 0$, show that $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$. Then apply it to X to bound $\mathbb{P}(X \geq 21)$.

[3 marks]

- (b) For a general random variable W with mean μ and variance σ^2 , derive the Chebyshev inequality $\mathbb{P}(|W - \mu| \geq k) \leq \sigma^2/k^2$ from Markov's inequality. Then apply it to bound $\mathbb{P}(|X - 12| \geq 6)$.

[2 marks]

- (c) Now suppose $X \sim \mathcal{N}(12, 9)$. Compute the exact value of $\mathbb{P}(|X - 12| \geq 6)$ and compare to the bound from (b).

[5 marks]

Solution:

- (a) Proof: Since $Y \geq 0$, we have $Y \geq a \cdot \mathbf{1}_{Y \geq a}$ (the indicator equals 1 when $Y \geq a$, and in that case $Y \geq a$; when the indicator is 0, $a \cdot 0 = 0 \leq Y$). Taking expectations: $\mathbb{E}[Y] \geq a \mathbb{E}[\mathbf{1}_{Y \geq a}] = a \mathbb{P}(Y \geq a)$. Dividing by $a > 0$ gives $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$. $\square$

Application: $\mathbb{P}(X \geq 21) \leq \frac{\mathbb{E}[X]}{21} = \frac{12}{21} = \frac{4}{7} \approx 0.571$.

- (b) Let $Y = (W - \mu)^2 \geq 0$. Then $\{|W - \mu| \geq k\} = \{(W - \mu)^2 \geq k^2\}$. Applying Markov: $\mathbb{P}(|W - \mu| \geq k) = \mathbb{P}(Y \geq k^2) \leq \frac{\mathbb{E}[Y]}{k^2} = \frac{\text{Var}(W)}{k^2}$.

Application: $\mathbb{P}(|X - 12| \geq 6) \leq \frac{9}{36} = \frac{1}{4}$.

- (c) Let $Z = (X - 12)/3 \sim \mathcal{N}(0, 1)$.

$\mathbb{P}(|X - 12| \geq 6) = \mathbb{P}(|Z| \geq 2) = 2(1 - \Phi(2)) \approx 2 \times 0.0228 = 0.0456$.

The Chebyshev bound of $1/4 = 0.25$ is much more conservative than the exact value of 0.0456.

Question 4. [10 marks]

A continuous random variable X has PDF $f(x) = cx^3$ for $0 \leq x \leq 2$, and $f(x) = 0$ otherwise.

- (a) Find the normalizing constant c and compute the moment generating function $M_X(t) = \mathbb{E}[e^{tX}]$ as an integral (you may leave it in integral form for general t , but evaluate $M_X(0)$ to verify your answer).

[3 marks]

- (b) Find the CDF $F(x)$ and use it to compute $\mathbb{P}(1 < X < 1.5)$.

[2 marks]

- (c) Compute $\mathbb{E}[X]$ and $\mathbb{V}\text{ar}(X)$.

[2 marks]

- (d) Let $W = X^2$. Find $\mathbb{E}[W]$ and determine the PDF of W , stating its support clearly.

[3 marks]

Solution:

(a) $\int_0^2 cx^3 dx = c \cdot \frac{x^4}{4} \Big|_0^2 = c \cdot \frac{16}{4} = 4c = 1$, so $c = 1/4$.

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_0^2 e^{tx} \cdot \frac{x^3}{4} dx = \frac{1}{4} \int_0^2 x^3 e^{tx} dx.$$

Verification: $M_X(0) = \frac{1}{4} \int_0^2 x^3 dx = \frac{1}{4} \cdot 4 = 1$. ✓

(b) For $0 \leq x \leq 2$: $F(x) = \int_0^x \frac{t^3}{4} dt = \frac{x^4}{16}$.

$F(x) = 0$ for $x < 0$, and $F(x) = 1$ for $x > 2$.

$$\mathbb{P}(1 < X < 1.5) = F(1.5) - F(1) = \frac{1.5^4}{16} - \frac{1}{16} = \frac{5.0625 - 1}{16} = \frac{4.0625}{16} = \frac{65}{256} \approx 0.2539.$$

(c) $\mathbb{E}[X] = \int_0^2 x \cdot \frac{x^3}{4} dx = \frac{1}{4} \cdot \frac{x^5}{5} \Big|_0^2 = \frac{1}{4} \cdot \frac{32}{5} = \frac{8}{5} = 1.6$.

$$\mathbb{E}[X^2] = \int_0^2 x^2 \cdot \frac{x^3}{4} dx = \frac{1}{4} \cdot \frac{x^6}{6} \Big|_0^2 = \frac{1}{4} \cdot \frac{64}{6} = \frac{8}{3}.$$

$$\mathbb{V}\text{ar}(X) = \frac{8}{3} - \left(\frac{8}{5}\right)^2 = \frac{8}{3} - \frac{64}{25} = \frac{200 - 192}{75} = \frac{8}{75} \approx 0.1067.$$

(d) $\mathbb{E}[W] = \mathbb{E}[X^2] = \frac{8}{3}$ (computed above).

To find the PDF of $W = X^2$: since $X \in [0, 2]$ and $g(x) = x^2$ is strictly increasing on $[0, 2]$, we use the transformation formula. Let $w = x^2$, so $x = \sqrt{w}$ and $\frac{dx}{dw} = \frac{1}{2\sqrt{w}}$.

$$f_W(w) = f_X(\sqrt{w}) \cdot \left| \frac{dx}{dw} \right| = \frac{(\sqrt{w})^3}{4} \cdot \frac{1}{2\sqrt{w}} = \frac{w^{3/2}}{4} \cdot \frac{1}{2\sqrt{w}} = \frac{w}{8}$$

for $0 \leq w \leq 4$, and $f_W(w) = 0$ otherwise.

Verification: $\int_0^4 \frac{w}{8} \, dw = \frac{1}{8} \cdot \frac{w^2}{2} \Big|_0^4 = \frac{16}{16} = 1. \checkmark$

Double-check: $\mathbb{E}[W] = \int_0^4 w \cdot \frac{w}{8} \, dw = \frac{1}{8} \cdot \frac{w^3}{3} \Big|_0^4 = \frac{64}{24} = \frac{8}{3}. \checkmark$

Question 5. [10 marks]

Consider the following Python program:

```
def simulate():
    u1 = random()
    if u1 < 0.4:
        u2 = random()
        if u2 < 0.5:
            return 0
        else:
            return 1
    else:
        u2 = random()
        count = 0
        while u2 >= 0.3 and count < 3:
            count += 1
            u2 = random()
        return count
```

Let X be the value returned by `simulate()`.

- (a) Carefully determine the PMF of X . (*Hint*: consider cases A and B separately, noting that case B generates a truncated geometric-like count.)

[3 marks]

- (b) Compute $\mathbb{E}[X]$ and $\mathbb{Var}(X)$.

[3 marks]

- (c) Find the probability generating function $G_X(s) = \mathbb{E}[s^X]$ of X , and confirm that $G'_X(1) = \mathbb{E}[X]$.

[4 marks]

Note about Python: `random()` generates a $\mathcal{U}(0, 1)$ random variable.

Solution:

- (a) **Case A** (probability 0.4): returns 0 with probability 0.5, returns 1 with probability 0.5.

Case B (probability 0.6): The variable `count` starts at 0 and increases by 1 each iteration of the while loop. The loop continues while $u2 \geq 0.3$ and $\text{count} < 3$. Each new $u2$ is drawn independently, so at each iteration, the loop exits with probability 0.3 or continues with probability 0.7.

- $\mathbb{P}(\text{return } 0 \mid B)$: first $u2 < 0.3$, probability = 0.3.
- $\mathbb{P}(\text{return } 1 \mid B)$: first $u2 \geq 0.3$, then new $u2 < 0.3$, probability = $0.7 \times 0.3 = 0.21$.
- $\mathbb{P}(\text{return } 2 \mid B)$: two continues then exit, probability = $0.7^2 \times 0.3 = 0.147$.
- $\mathbb{P}(\text{return } 3 \mid B)$: three continues (loop exits since $\text{count} = 3$), probability = $0.7^3 = 0.343$.

Combining via the law of total probability:

$$\mathbb{P}(X = 0) = 0.4(0.5) + 0.6(0.3) = 0.20 + 0.18 = 0.38,$$

$$\mathbb{P}(X = 1) = 0.4(0.5) + 0.6(0.21) = 0.20 + 0.126 = 0.326,$$

$$\mathbb{P}(X = 2) = 0.6(0.147) = 0.0882,$$

$$\mathbb{P}(X = 3) = 0.6(0.343) = 0.2058.$$

Check: $0.38 + 0.326 + 0.0882 + 0.2058 = 1$. ✓

$$(b) \mathbb{E}[X] = 0(0.38) + 1(0.326) + 2(0.0882) + 3(0.2058) = 0 + 0.326 + 0.1764 + 0.6174 = 1.1198.$$

$$\mathbb{E}[X^2] = 0(0.38) + 1(0.326) + 4(0.0882) + 9(0.2058) = 0 + 0.326 + 0.3528 + 1.8522 = 2.5310.$$

$$\mathbb{V}\text{ar}(X) = 2.5310 - 1.1198^2 = 2.5310 - 1.2540 = 1.2770.$$

$$(c) G_X(s) = 0.38 + 0.326 s + 0.0882 s^2 + 0.2058 s^3.$$

$$G'_X(s) = 0.326 + 0.1764 s + 0.6174 s^2.$$

$$G'_X(1) = 0.326 + 0.1764 + 0.6174 = 1.1198 = \mathbb{E}[X]. \quad \checkmark$$

Question 6. [10 marks]

Let $X \sim \text{Exp}(1)$ and $Y \sim \text{Exp}(3)$ be independent.

- (a) Write the PDFs $f_X(x)$ and $f_Y(y)$, and compute $\mathbb{E}[X + Y]$ and $\mathbb{V}\text{ar}(X + Y)$ using properties of expectation and variance.

[2 marks]

- (b) Using the convolution formula, derive the PDF of $Z = X + Y$ for $z > 0$.

[2 marks]

- (c) Find the MGF $M_Z(t)$ of Z and use it to compute $\mathbb{E}[Z]$. Confirm it matches (a).

[2 marks]

- (d) Find the CDF $F_Z(z)$ and use it to compute $\mathbb{E}[Z]$ via $\int_0^\infty (1 - F_Z(z)) dz$. Confirm it matches (a) and (c).

[2 marks]

- (e) Define $W = X - Y$. Using the transformation $(X, Y) \rightarrow (Z, W)$ where $Z = X + Y$, find the joint PDF $f_{Z,W}(z, w)$ and state its support. Are Z and W independent? Find the conditional PDF $f_{W|Z}(w|z)$.

[2 marks]

Solution:

- (a) $f_X(x) = e^{-x}$, $x \geq 0$. $f_Y(y) = 3e^{-3y}$, $y \geq 0$.

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] = 1 + \frac{1}{3} = \frac{4}{3}.$$

$$\mathbb{V}\text{ar}(X + Y) = \mathbb{V}\text{ar}(X) + \mathbb{V}\text{ar}(Y) = 1 + \frac{1}{9} = \frac{10}{9}.$$

$$\begin{aligned} \text{(b)} \quad f_Z(z) &= \int_0^z f_X(x) f_Y(z-x) dx = \int_0^z e^{-x} \cdot 3e^{-3(z-x)} dx \\ &= 3e^{-3z} \int_0^z e^{-x+3x} dx = 3e^{-3z} \int_0^z e^{2x} dx = 3e^{-3z} \cdot \frac{e^{2z} - 1}{2} \\ &= \frac{3}{2}(e^{-z} - e^{-3z}), \quad z > 0. \end{aligned}$$

- (c) Since X and Y are independent: $M_Z(t) = M_X(t) M_Y(t) = \frac{1}{1-t} \cdot \frac{3}{3-t} = \frac{3}{(1-t)(3-t)}$.

By the quotient rule with numerator $u = 3$ and denominator $v = (1-t)(3-t) = 3 - 4t + t^2$:

$$v' = -4 + 2t, \text{ so } M'_Z(t) = \frac{0 \cdot v - 3 \cdot (-4 + 2t)}{v^2} = \frac{3(4 - 2t)}{((1-t)(3-t))^2}.$$

$$\mathbb{E}[Z] = M'_Z(0) = \frac{3 \cdot 4}{(1 \cdot 3)^2} = \frac{12}{9} = \frac{4}{3}. \quad \checkmark$$

(d) The CDF of Z : for $z > 0$,

$$F_Z(z) = \int_0^z \frac{3}{2}(e^{-t} - e^{-3t}) dt = \frac{3}{2} \left[-e^{-t} + \frac{e^{-3t}}{3} \right]_0^z = \frac{3}{2} (1 - e^{-z}) - \frac{1}{2} (1 - e^{-3z}) = 1 - \frac{3}{2}e^{-z} + \frac{1}{2}e^{-3z}.$$

Then $1 - F_Z(z) = \frac{3}{2}e^{-z} - \frac{1}{2}e^{-3z}$, and

$$\mathbb{E}[Z] = \int_0^\infty (1 - F_Z(z)) dz = \frac{3}{2} \int_0^\infty e^{-z} dz - \frac{1}{2} \int_0^\infty e^{-3z} dz = \frac{3}{2} - \frac{1}{6} = \frac{4}{3}. \checkmark$$

(e) Inverse transformation: $X = (Z + W)/2$, $Y = (Z - W)/2$. The Jacobian matrix is

$$\frac{\partial(X, Y)}{\partial(Z, W)} = \begin{pmatrix} \partial X / \partial Z & \partial X / \partial W \\ \partial Y / \partial Z & \partial Y / \partial W \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}.$$

Determinant: $(1/2)(-1/2) - (1/2)(1/2) = -1/2$, so $|J| = 1/2$.

Joint PDF: $f_{Z,W}(z, w) = f_X\left(\frac{z+w}{2}\right) f_Y\left(\frac{z-w}{2}\right) \cdot |J|$

$$= e^{-(z+w)/2} \cdot 3e^{-3(z-w)/2} \cdot \frac{1}{2} = \frac{3}{2} e^{-(z+w)/2 - 3(z-w)/2} = \frac{3}{2} e^{-2z+w}$$

for $z > 0$ and $-z < w < z$ (from $X = (z + w)/2 \geq 0$ and $Y = (z - w)/2 \geq 0$).

Z and W are **not independent**: the joint PDF does not factor as $g(z) \cdot h(w)$ over a rectangular support (the support $\{-z < w < z\}$ depends on z).

$$\text{Conditional: } f_{W|Z}(w|z) = \frac{f_{Z,W}(z, w)}{f_Z(z)} = \frac{(3/2)e^{-2z+w}}{(3/2)(e^{-z} - e^{-3z})} = \frac{e^w}{e^z - e^{-z}}, \quad -z < w < z.$$

Table of Distributions

Discrete Distributions $(q = 1 - p)$

Distribution	PMF $p(x)$	Support	$\mathbb{E}X$	$\mathbb{V}\text{ar}(X)$	PGF $G_X(z)$
$\text{DU}(\{1, \dots, N\})$	$\frac{1}{N}$	$\{1, \dots, N\}$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$	$\frac{z(1-z^N)}{N(1-z)}$
$\text{Ber}(p)$	$p^x q^{1-x}$	$\{0, 1\}$	p	pq	$q + pz$
$\text{Bin}(n, p)$	$\binom{n}{x} p^x q^{n-x}$	$\{0, \dots, n\}$	np	npq	$(q + pz)^n$
$\text{Geom}(p)$	$p q^{x-1}$	$\{1, 2, \dots\}$	$\frac{1}{p}$	$\frac{q}{p^2}$	$\frac{pz}{1-qz}$
$\text{NegBin}(r, p)$	$\binom{x-1}{r-1} p^r q^{x-r}$	$\{r, r+1, \dots\}$	$\frac{r}{p}$	$\frac{rq}{p^2}$	$\left(\frac{pz}{1-qz}\right)^r$
$\text{Poi}(\lambda)$	$\frac{e^{-\lambda} \lambda^x}{x!}$	$\{0, 1, 2, \dots\}$	λ	λ	$e^{\lambda(z-1)}$
$\text{Hyp}(N, K, n)$	$\frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$	$\{\max(0, n+K-N), \dots, \min(n, K)\}$	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{N^2(N-1)}$	—

Continuous Distributions

Distribution	PDF $f(x)$	Support	$\mathbb{E}X$	$\mathbb{V}\text{ar}(X)$	MGF $M_X(t)$
$\mathcal{U}(a, b)$	$\frac{1}{b-a}$	$[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$
$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$[0, \infty)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$
$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	$\mathbb{R}$	μ	σ^2	$e^{\mu t + \sigma^2 t^2/2}$
$\text{Gamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$	$(0, \infty)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda - t}\right)^\alpha$
$\chi^2(n)$	$\frac{x^{n/2-1} e^{-x/2}}{2^{n/2} \Gamma(n/2)}$	$(0, \infty)$	n	$2n$	$(1-2t)^{-n/2}$
$\text{Beta}(\alpha, \beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$(0, 1)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—
$\text{Pareto}(\alpha)$	$\alpha x^{-(\alpha+1)}$	$[1, \infty)$	$\frac{\alpha}{\alpha-1} \ (\alpha > 1)$	$\frac{\alpha}{(\alpha-1)^2(\alpha-2)} \ (\alpha > 2)$	—